PolicyGuard AI: Final Report

TEAM011 · Joydev Pain (2025MB26105)
MBA ZG583 Management of AI Products

All data, rules, companies and people in this project are fictional. Business figures are planning assumptions.

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02 · Product Strategy and Opportunity Framing

Product vision

> **Every policy folder is audit-ready before anyone asks, and every renewal decision starts from complete evidence.** People make the decisions. AI makes sure they see the gaps in time.

Long term, PolicyGuard AI can become a compliance layer across business lines, covering bulk monitoring and remediation tracking. That future must be earned through a proven single-folder workflow.

Jobs to be done (JTBD)

Core job (underwriting operations analyst)

> When **I need to check a policy folder for audit or renewal**, I want to **see what is missing, wrong or inconsistent, with evidence**, so that I can **fix real gaps early without reading every page**.

Related jobs

Who	Situation	Motivation	Outcome
Compliance officer	Preparing for a third-party audit	See audit readiness before the auditor does	Fewer late findings
Renewal handler	A policy is approaching renewal	Know whether documents support renewal	Faster renewal with fewer gaps
Underwriting manager	Tracking team workload	Apply one standard across analysts	Consistent review quality
New analyst	Learning what a compliant folder looks like	Understand why something fails	Faster onboarding through linked rules

Emotional job: "I don't want an auditor to find something that I signed off."

Opportunity framing

Why the problem matters - Compliance gaps found late create regulatory and reputational exposure. - Manual checking uses analyst time that could go to risk decisions. - Renewals can be delayed or processed on incomplete evidence. - The same issues recur across folders, so remediation effort is duplicated.

Why current alternatives are weak - File-system scripts match names, not meaning. A file named `App` was treated as a missing Application in my baseline. - Repositories store documents but do not assess them. - Manual review is slow, and depends on who performs it.

Why AI can help, and why now - Multimodal LLMs can read both digital PDFs and scanned pages. - Structured output makes it practical to require a quote, page and rule for each finding. - API access removes the need to train a model. My illustrative estimate is about US$0.025 per folder (see Part 06).

AI opportunity matrix (the compliance and renewal workflow)

Each candidate use of AI was scored for value, feasibility and the cost of being wrong. **"AI approves the renewal" was rejected on purpose.**

AI opportunity	Business value	Feasibility (data + tech)	Risk if wrong	Priority
Identify document type from content	High	High	Low: rules and people still check	**MVP**
Extract key fields with evidence	High	High on clean documents; unproven on messy scans	Medium	**MVP**
Check content against retrieved rules	High	Medium-high: needs evidence checks	Medium	**MVP**
Detect cross-document conflicts	High	Medium-high	Medium	**MVP**
Draft a renewal rationale from fixed facts	Medium	High	Low: rejected if citations fail	**MVP**
Bulk scan of all folders	High	Medium	Medium	Phase 3
Recurring-issue analysis across folders	Medium	Medium	Low	Phase 3
Automatic remediation of documents	Medium	Low	High	Not now
Approve renewals automatically	High on paper	Low	**Very high**: regulatory and customer impact	**Rejected**
Risk pricing from folder contents	High	Low	Very high: fairness and regulation	**Rejected**

Feature vs platform

The *Amazon vs ChatGPT* case frames this choice. PolicyGuard AI begins as a **feature of one workflow**, single-folder compliance review, and not as a general "chat with your policies" assistant. One workflow means one user, one decision moment and metrics that can be tested. Bulk monitoring can follow once single-folder accuracy holds on real documents.

Build vs buy vs partner/API

Option	Pros	Cons	Verdict
Buy a document-management or generic AI search tool	Fast; may already be licensed	Rarely encodes the insurer's rules; little control over evidence, routing or review capture	**Use as a comparison if already licensed**
API-first RAG (LLM and embeddings API, with the insurer's own rules, checks and workflow)	Fast to build; controls evidence, rules and logging; can be evaluated properly	Needs an approved enterprise agreement; depends on a provider	**Recommended**
Partner with a document-AI or insurtech vendor	Ready extraction for insurance forms; faster start on scans	The vendor's roadmap controls the rule logic; per-page fees; policyholder data passes through one more party	**Revisit in Phase 1** if scans need a dedicated OCR service
Build or fine-tune a model	Full control	Expensive and slow; does not solve evidence or workflow	Not justified

Recommended path: API-first RAG, with workflow and compliance logic that the insurer owns. **Why:** the value lies in the organization's rules and its review process, not in owning a model. **Main trade-offs:** provider dependency is limited by a single AI connection module, `backend/llm_client.py`. Policyholder data leaving the network is a serious issue. It requires enterprise terms and privacy approval, or an in-house deployment if these are not met.

First wedge

Commercial property policies, one audit or renewal queue, 6–8 analysts, 12 weeks.

Why this wedge: - The prototype's required documents (application, risk assessment, inspection report, signed disclosure) match this line. - Inspection reports and risk grades give meaningful content checks. - Commercial data has less personal and health information than health insurance, which limits privacy risk in a first pilot. - One business line keeps rule encoding and labeling manageable.

The assumptions behind this wedge need confirmation in Phase 0.

Course case lenses

Case	What is similar	What is different	Lesson that transfers	What I would do differently
Google SGE / Yellow.ai (assistant UX)	AI output is shown on top of source content	Reviewers here must defend each finding to an auditor	People trust AI answers they can check against the source	Show the rule, quote and confidence on every finding, and allow the tool to decline to score
Netflix (metrics and evaluation)	Offline metrics only matter if they link to business outcomes	Too few users here for routine A/B tests	Tie model metrics to the outcome that pays for the product	Use a metric tree from defect recall to audit findings, and a shadow-mode pilot in place of A/B tests
Amazon vs ChatGPT (feature vs platform)	A general assistant is a tempting thing to build	This product has one defined workflow	A platform has to be earned through a working feature	Stay a single-workflow feature until real-document accuracy is proven, and leave out chat
iMerit (data operations and labeling)	Quality depends on trusted labels	Only compliance experts can label these findings	Treat labeling as an operation with a real budget	Budget for a 300-folder golden set labeled independently by two experts
Instacart / Tesla (lifecycle and drift)	Conditions change after launch	Drift here comes from new forms, rules and model versions, not from user behavior	Plan for drift before launch	Version the rulebook, pin the model, and rerun the evaluation before any change
Apple Card / Bias in Lending (ethics and governance)	Automated output can affect customers unequally	This product does not decide eligibility or price	Harm can come through a process, even without a decision	Keep AI out of approval and pricing, and check whether flags cluster by branch, era or segment

03 · PM Artifact Pack

The core PM documents for PolicyGuard AI follow. Section 9 explains what being an AI product changed in each one. The full PRD is in Appendix/PRD.

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1. Problem statement

> **Underwriting operations analysts** (initially commercial property) **cannot check policy folders for structure, naming and content quickly or consistently enough**. **This causes** late discovery of compliance gaps, delayed or poorly evidenced renewals, and results that depend on the reviewer. **The decisions being improved** are the reviewer's *confirm / reject / escalate* decision on each finding, and the underwriter's renewal decision.

Out of scope: automatic compliance sign-off, automatic renewal approval, pricing, or document remediation.

2. Product vision

Every policy folder is audit-ready before anyone asks, and people make every compliance and renewal decision on complete evidence.

3. Personas (AI-aware)

The personas below are illustrative and have not been validated through user interviews.

Persona	Context	Wants from the tool	What an AI mistake costs them	What they need before trusting it
Meera, underwriting operations analyst (3 yrs)	Reviews folders daily across legacy naming styles	Fast, specific findings	**High** if a missed defect goes to audit; medium if false alarms waste time	Must see the document, rule and quote
Arjun, compliance officer (12 yrs)	Answers to auditors and regulators	Consistent reports and a clear trail	**Very high**: a wrong sign-off creates regulatory exposure	Needs proof of human review
Farah, underwriting manager	Owns renewal decisions and team workload	Renewal suggestions with reasons	High: renewal on incomplete evidence	Must see why a renewal was held or referred

4. PRD summary

Item	Content
User problem	Folder compliance checks are slow, late and inconsistent
Core workflow	Choose folder → AI reads documents → rules retrieved and checked → score, evidence and renewal suggestion → human decision on each finding → sign-off → audit log and report
AI role	(1) Read and classify documents; (2) extract fields with quotes; (3) assess content against retrieved rules; (4) find conflicts across documents; (5) draft a rationale from fixed facts. The AI **approves nothing**.
Data assumptions	Real folders can be accessed lawfully. Rules can be written as clear statements. Scans are legible enough. *All to be verified in Phase 0.*
Key risks	Missed defects; false or fabricated evidence; over-reliance; policyholder data exposure; inconsistent results between runs
Fallback logic	Any document cannot be analysed → Not Scored, no recommendation. Rule not among those retrieved → finding removed. AI rationale fails citation check → template rationale. Signature rules always run, even if retrieval misses them.

Item	Content
Non-functional	One folder analysed in under 2 minutes; access control; no provider retention or training; pinned model versions; all decisions logged

5. Feature prioritization (RICE)

Reach counts folders per quarter in the pilot queue (3,000, based on 12,000 a year). Impact uses a 0.25–3 scale. Confidence shows how well the impact is evidenced. Effort is in person-weeks.

Feature	Reach	Impact	Confidence	Effort	RICE	Decision
Fail-closed handling	3,000	3	100%	1	**9,000**	MVP
Structure and naming checks	3,000	2	100%	1	**6,000**	MVP
Human decision, sign-off and audit log	3,000	2	90%	1	**5,400**	MVP
Cross-document checks	3,000	3	80%	2	**3,600**	MVP
Evidence and confidence bands	3,000	3	80%	2	**3,600**	MVP
AI document reading and extraction	3,000	3	80%	3	**2,400**	MVP
RAG rule retrieval and content assessment	3,000	2	60%	3	**1,200**	MVP
Renewal suggestion	1,000	2	70%	2	**700**	MVP
Recurring-issue dashboard	3,000	1	50%	4	**375**	Phase 3
Bulk folder scan	3,000	2	50%	8	**375**	Phase 3
Chat over policy documents	3,000	0.5	40%	4	**150**	Later
Auto-remediation	1,000	2	20%	8	**50**	Not now

Scores I overrode, and why. RAG ranks below several cheaper features, and AI reading ranks below the plain-rule checks. Both stay in the MVP because they carry the AI value hypothesis: without them, this would be the same script it is meant to beat. Bulk scan has real demand, but I delayed it because an unproven model run across thousands of folders multiplies its errors. Renewal has a lower reach because only part of the queue is at renewal.

6. MVP definition

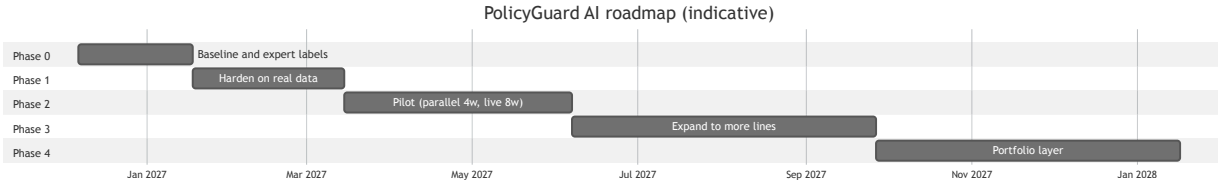
The claim the MVP has to test: *reviewers find real compliance gaps faster when AI shows linked evidence, and the system withholds a result when it cannot support one.*

In the MVP	Out of the MVP
One folder at a time, sample or upload	Bulk scanning and scheduling
Structure, naming, missing and duplicate checks	Direct integration with policy administration systems
AI reading with rule, quote, page and confidence	Word document analysis
RAG over a rulebook	Real insurer or regulatory rules
Cross-document checks	Auto-remediation
Audit-readiness score and fail-closed status	Claims history and pricing inputs for renewal
Renewal suggestion with rationale	Automatic renewal
Human decisions, sign-off, audit log, report	Single sign-on and role-based access

The MVP succeeds when the pilot targets in section 8 are met.

7. Roadmap with proof gates

Each phase has to earn the next. Work moves on only when the gate in the table below is met, so **staged learning and proof gates** set the pace, not the calendar.



Phase	What it must achieve	Gate to the next phase
0. Baseline (6 wks)	Establish how long reviews take now, and write the real rules down	Time study of 30 folders; approved rulebook; privacy approval for test data; 300 folders labeled by 2 experts (agreement ≥ 0.7)
1. Harden on real data (8 wks)	Show that reading works on real documents	All offline gates in Part 05: critical recall ≥ 95%, overall recall ≥ 85%, precision ≥ 70%, 0 fabricated evidence in 100 audited findings, repeatability ≥ 95%, retrieval recall ≥ 98%, security and privacy sign-off
2. Pilot (12 wks)	Test workflow value and trust. Analysts first run the tool in parallel with normal work (**shadow mode**), then use it live	≥ 40% less analyst time; ≥ 70% useful; rejection rate 10–40%; no serious incident
3. Expand (16 wks)	Add other lines, bulk scan and a recurring-issue view	Metrics hold across lines; a funded owner; run cost within budget
4. Portfolio	Shared compliance layer across portfolios	Needs its own business case

Rollback rule. In any phase, a fabricated finding in an audit submission, a critical miss traced to the tool, or a data breach sends the product back to shadow mode.

8. Success metrics

Type	Metric	Pilot target	Why it matters
Product	% of in-scope folders reviewed with the tool	≥ 80%	Adoption comes first
Product	Median analyst time per folder	−40% against the Phase 0 time study	This is the efficiency claim
Business	Time from gap arising to detection	Measured and falling	The core "catch it early" claim
Business	Findings first raised by third-party audit	Trend down	The risk-reduction claim
Business	Renewals delayed by document gaps	Trend down	Renewal value
Model	Recall on critical defects	≥ 95%	A missed critical defect is the worst failure
Model	Precision	≥ 70%	Too many false alarms destroy trust
Trust / guardrail	Fabricated evidence in audited findings	**0**	One can undermine an audit

Type	Metric	Pilot target	Why it matters
Trust / guardrail	Repeatability across runs	≥ 95%	Milestone 1 promised consistency
Trust / guardrail	False passes on failed analysis	**0**	The system must not hide failure
Trust / guardrail	Rejection rate	10–40%	Above 40% blocks launch; 5–10% needs investigation; below 5% suggests people may be accepting findings unread, and blocks launch unless outcomes support it
Trust / guardrail	"Useful" rating	≥ 70%	Reviewer trust

The metric tree is in Appendix/Metrics.

9. What being an AI product changed

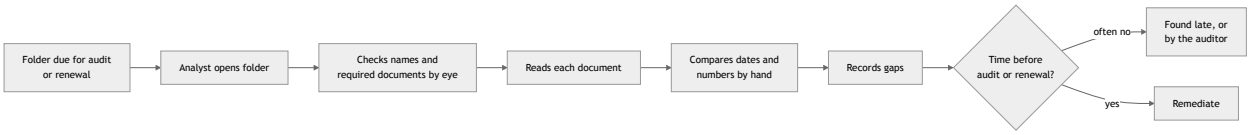
Artifact	Conventional version	PolicyGuard AI version
Problem statement	"Analysts need a better checklist"	States the **decisions** being improved and what a wrong one costs
Persona / user story	Goals and pain points	Each persona shows what an AI mistake costs them and what **uncertainty and trust** they need handled
PRD	Features and acceptance criteria	Also sets **data assumptions**, **limits on the AI**, **fallback logic** and **risks**
Prioritization	RICE order is followed	RICE is an input: overrides are explained, and fail-closed behavior is not negotiable
MVP	Fewest features that work	Fewest features that test both **value** and **trust**
Roadmap	Features by quarter	**Numeric gates**, shadow mode and a rollback rule
Metrics	Usage and satisfaction	**Model metrics linked to outcomes**, with **guardrails** and an acceptable rejection range
Launch checklist	QA and training complete	Adds review, rollback and escalation items, shown in the launch checklist in 07
Pricing / cost note	License cost	Includes **inference cost per folder** and the people cost of ongoing review
Review memo	"Build it"	Keeps **evidence** (synthetic test results) apart from **assumptions** (time saved), and concludes **pilot, not scale**

04 · UX, Workflow and Trust Design

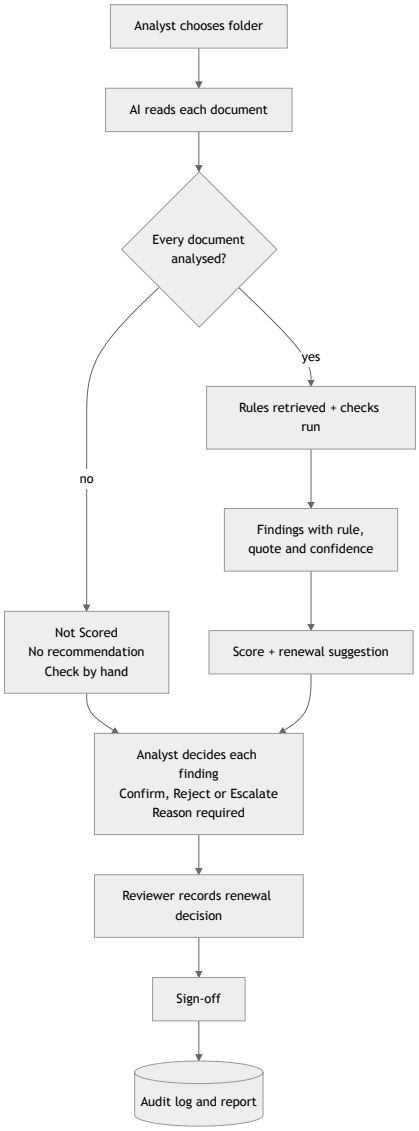
UX principle: PolicyGuard AI behaves like a **careful junior auditor who shows the evidence and leaves the decision to you**. AI output is probabilistic, so the interface centers on evidence, confidence and the human decision, not on a verdict.

Main workflow

Today (as-is)



With PolicyGuard AI (to-be)



Key user journey: Meera reviews folder POL200007

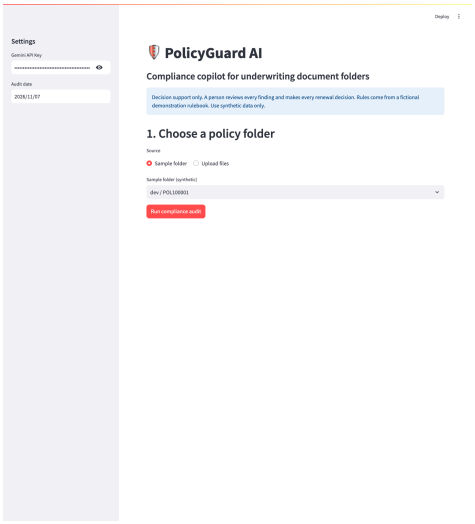
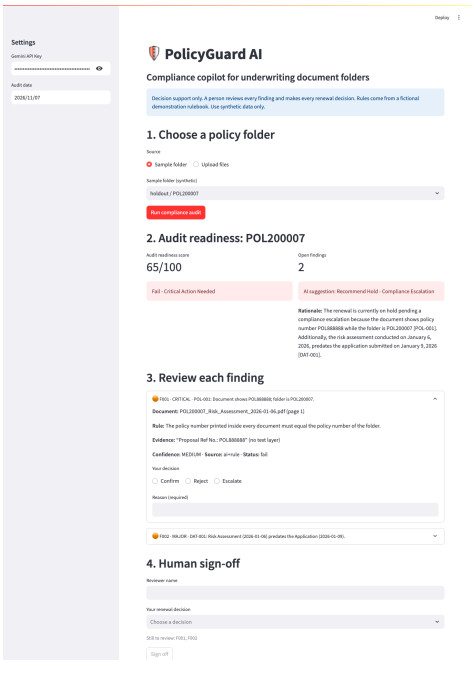
1. Meera selects the folder and runs the audit. It is a scanned folder with legacy labels such as "Proposal Ref No.". The audit takes about 40 seconds.

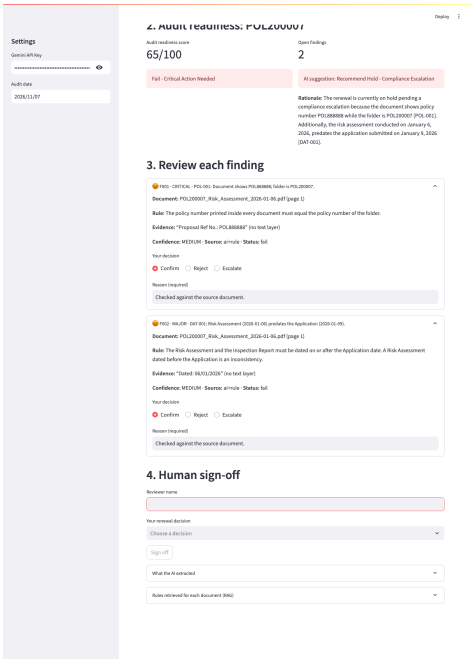
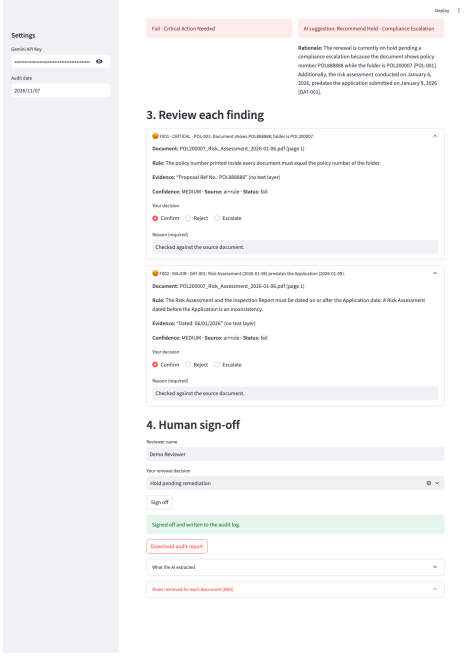
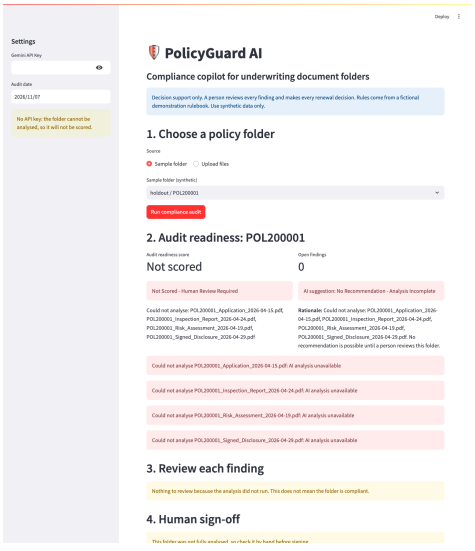
2. The tool shows **65/100, Fail – Critical Action Needed**, with an AI suggestion of **Recommend Hold – Compliance Escalation**.
3. Finding F001 is **critical**. The risk assessment shows POL888888, but the folder is POL200007. The evidence is "Proposal Ref No.: POL888888", with the note "no text layer".
4. Finding F002 is **major**. The risk assessment (6 Jan 2026) predates the application (9 Jan 2026).
5. Both are **MEDIUM confidence**, because a scan cannot be automatically matched to a quote. Meera opens the page and confirms each finding with a reason.
6. Sign-off stays disabled until she has decided each finding and chosen a renewal decision. She records "Hold pending remediation", signs off and downloads the report.
7. **What changed compared with a file-name script?** In this folder, the script finds nothing, because all the file names are valid. The wrong policy number and the date conflict are visible only inside the documents.

This journey is shown in Screenshot 02 and Screenshot 04.

Screens / wireframes

The working prototype serves as the wireframe. All screens use synthetic data.

Screen	Purpose
	**Choose a folder**: sample or upload, with the "decision support only" notice
	**Audit result**: score, status, renewal suggestion and rationale with rule IDs

Screen	Purpose
	**Human review**: rule, evidence and confidence, with a required decision and reason
	**Sign-off**: reviewer, renewal decision, audit log and report
	**Fail closed**: not scored, no recommendation, and a clear warning

Layout notes are in Appendix/Wireframes.

Trust and explainability

Question	Answer
What does the user need to know?	What is wrong, where, under which rule, how sure the system is, and what is theirs to decide
What must be visible?	Severity; rule ID and text; document and page; quoted evidence and whether it was verified; confidence band; "decision support only" notice
What can be shown on request?	All extracted fields; the rules retrieved for each document
What should be hidden?	Raw embeddings and internal technical details. The API key field is masked and must not exist in production.
What the AI never shows	"Approved"; an automatic sign-off; any renewal outcome presented as final

Uncertainty handling

State	Trigger	Behavior
HIGH	Quote found in the document text, and the model reports high confidence	Normal review. Applies also to plain structure checks.
MEDIUM	Quote found with medium model confidence, or a scan where the quote cannot be checked	Review prompted, with "no text layer" shown
LOW	Quote not found in the document, or low model confidence	Flagged in red for careful checking
Unclear	AI or a rule cannot determine the answer	Finding stays open for a person, with no points deducted
Not Scored	Any document, or the folder review, fails	No score, no recommendation, and a warning that this does not mean the folder is compliant
Invalid citation	AI cites a rule that was not retrieved	Statement removed and counted
Rationale fails check	AI text cites unknown rules or says "approved"	Template rationale shown instead

Every test finding fell in the HIGH (24) or MEDIUM (7) band, so **LOW behavior has been tested only in code, not seen in a live run**. Confidence bands also need calibration on real data in Phase 1 (see Part 05).

Human-in-the-loop

Point	Who	What they do
Every finding	Analyst	Confirms, rejects or escalates. **A reason is required.**
Every folder	Reviewer	Records a renewal decision and signs off. **No option is pre-selected.**
Escalation	Compliance or underwriting SME	Decides ambiguous or high-risk cases
Audit submission	Compliance officer	Approves the report sent to the third-party auditor
Monthly	Rulebook owner	Reviews rejected findings and escalations, and updates rules
Quarterly	Product owner + compliance	Reruns the evaluation and reviews drift

Designing against over-reliance: a passing folder still needs sign-off, each decision needs a reason, the renewal choice starts blank, and a very low rejection rate is investigated, not celebrated. The blank default exists because testing exposed the risk: the first upgraded version pre-selected "Approve renewal" on a folder that the AI suggested holding.

05 · Data, Model and Evaluation Strategy

1. Problem as an AI task

PolicyGuard AI is a **pipeline of steps**, not one model. AI is used only where reading or judgment is needed.

Step	Task	Output	Method in prototype	Pilot
1. Read and extract	**Multimodal document understanding**	Type, key fields and signature, each with quote and page	Gemini with a structured schema	Enterprise LLM; a dedicated OCR service if needed
2. Find rules	**Semantic retrieval (RAG)**	Relevant rules	Gemini embeddings + ChromaDB	Same, on the real rulebook, possibly hybrid
3. Assess content	**Grounded classification**	Pass, fail or unclear per rule	Gemini, limited to retrieved rules	Same, recalibrated
4. Check structure and consistency	**Rules**	Missing, duplicate, naming, date and policy-number findings	Python rules (deliberately not AI)	Same
5. Compare soft fields	**Cross-document reasoning**	Name, location or sum insured conflicts	Gemini	Same
6. Score and suggest	**Rules**	Score, status, renewal suggestion	Python rules (deliberately not AI)	Same, with owner-approved weights
7. Explain	**Grounded generation**	Plain-language rationale	Gemini, with citation check and template fallback	Same

Date comparison and scoring stay as rules on purpose. They need to be exact, repeatable and easy to explain to an auditor.

2. Data sources

Type	Source	Use	Notes
Internal	Policy document folders	Main input	Prototype uses synthetic folders only
Internal	Underwriting guidelines and compliance rules	Rulebook for RAG	Prototype uses 22 fictional rules; these are not real regulations
Internal	Past audit findings	Golden-set labels, recurring issues	Pilot only
Internal	Policy administration data	Renewal dates, policy status	Pilot only
User-generated	Reviewer decisions, reasons, sign-offs	Evaluation, rule tuning	Captured in the prototype's audit log
External	Regulatory publications (US state insurance regulators, NAIC guidance)	Requirements behind the rulebook	Compliance team interprets these; AI does not

3. Data assumptions (to verify in Phase 0)

#	Assumption	Check	If the check fails
A1	Real documents can be used lawfully for the pilot	Privacy and legal review	Mask data, or deploy in-house

#	Assumption	Check	If the check fails
A2	Compliance rules can be written as clear, testable statements	Workshop with compliance	Keep ambiguous rules for human checks only
A3	Most folders share a recognizable document set	Sample 100 folders	Add per-line taxonomies
A4	Legacy scans are readable	Sample scans by era	Add an OCR service; route poor scans to people
A5	Past audit findings can act as usable labels	Match 50 findings to folders	Label from scratch, with higher cost
A6	Signatures can be judged from images	Expert review of 100 real examples	Treat signatures as a human check

4. Labeling / ground truth

Prototype. Ground truth comes from planted defects. `eval/generate_folders.py` creates each folder from a known list of defects and records the expected rules and renewal outcome in `eval/ground_truth.json`.

Pilot. The question to answer is *"what would a senior compliance reviewer flag in this folder?"* That label will be created as follows:

1. **300 real folders**, sampled across eras, branches and quality.
2. **Two experts independently** label each folder against the approved rulebook.
3. **Agreement** measured with Cohen's kappa. Target ≥ 0.7. A third expert resolves disagreements.
4. **Evidence audit:** a reviewer checks 100 AI findings against the source pages.
5. **Repeatability test:** 50 folders are run three times each.
6. **Ongoing:** each month, rejected findings and escalations are reviewed, and useful ones join the golden set.

Effort: about 450 expert hours (300 folders × 45 minutes × 2 reviewers), which is included in the business case.

5. Baseline

A file-name-only script. It uses the same structure rules with no AI, standing in for what simple automation can do. The pilot adds the stronger baseline: **current manual review**, measured by the Phase 0 time study and by defects found.

6. Model / API choice

Component	Prototype choice	Why
LLM	`gemini-3.5-flash-lite`	Reads PDFs and images, fast, and available on the free tier
Embeddings	`gemini-embedding-001` (3,072 dimensions)	No training required
Vector store	ChromaDB, local	Real vector retrieval with simple setup
Fine-tuning	**No**	Does not fix evidence or grounding, and adds provider dependence
Guardrails	Required quotes; quote check; retrieved-rule check; always-on signature rules; fail-closed scoring; rationale check	Each targets a specific failure seen in testing or design

Why this model: the work started with `gemini-3.6-flash`, but its free tier allows only 20 requests a day, and the evaluation needed about 330. The change was for practical reasons, not after a quality comparison. **A model comparison is still needed in Phase 1.**

7. Evaluation framework

Metric type	Metric	Why it matters	Pilot threshold
Technical	**Recall on critical defects**	The worst error is a missed critical defect	≥ 95%
Technical	Overall recall	Coverage	≥ 85%
Technical	Precision	False alarms waste time and trust	≥ 70%
Technical	**Fabricated evidence**	Hard guardrail	**0**
Technical	Repeatability	Consistency was a Milestone 1 promise	≥ 95%
Technical	Rule retrieval recall	A missed rule is a check that never runs	≥ 98%
Technical	False pass on failed analysis	Must never hide failure	**0**
Technical	Time per folder	Usability	< 2 minutes
Business	Analyst time per folder	Efficiency	−40%
Business	Findings first raised by auditors	Risk reduction	Trend down
Trust	Usefulness rating	Adoption	≥ 70%
Trust	Rejection rate	Healthy skepticism	10–40%

8. Prototype evaluation results (synthetic data)

Run `python eval/evaluate.py dev` or `holdout` from `policyguard-ai/`. Full outputs are in eval/RESULTS.md (https://github.com/MAIP2026July/TEAM011) and the `results_*.json` files.

Test design

- **24 folders, 95 PDFs and 30 planted defects**, covering 19 of the 22 rules. The other 3 are renewal rules, which were tested through the renewal outcome.
- **Dev (12 folders, 14 defects):** standard wording, used for tuning.
- **Holdout (12 folders, 16 defects):** different labels and DD/MM/YYYY dates. It was run **once, after the code was frozen, with no tuning afterward**.
- Signatures are **drawn images**, not text, so the model must see them. Five folders contain only scanned pages.
- Each folder includes 0–3 defects. Three folders in each set are clean.
- Audit date fixed at 1 Nov 2026.

Detection

Set	System	Found	Missed	False alarms	Recall	Precision
Dev	File-name baseline	4 of 14	10	1	28.6%	80.0%
Dev	PolicyGuard AI	14 of 14	0	0	100%	100%
Holdout	File-name baseline	4 of 16	12	2	25.0%	66.7%
Holdout	**PolicyGuard AI**	**16 of 16**	**0**	**1**	**100%**	**94.1%**

Other holdout results

Measure	Result
Critical defects missed	0
Clean folders falsely flagged	0 of 3
Renewal suggestions matching expected outcome	12 of 12
Failed documents	0

Measure	Result
Citations rejected	0
Rule retrieval recall at 10	100%
Mean time per folder in the harness	110 s, including a deliberate wait between calls
Time in the live app	33–42 s for POL200007
AI calls per folder	About 14

How to read these results

1. **The holdout result is encouraging, but it is not proof.** With 16 of 16 found, the 95% confidence interval still allows true recall as low as about 79%, even on this easy data.
2. **The test favors the system.** One AI-assisted process wrote the rules, documents and defects. Pages are single, tidy and cleanly typed, and signatures are simple drawn lines. Real folders will be harder, so **real-world accuracy is likely to be lower**.
3. **The one false alarm is informative.** On scanned POL200007, the extraction step accepted the signature, but the rule step called it "a squiggle". Two steps of one system disagreed about one image. The finding was MEDIUM confidence, and a person would review it, but this is a real weakness in signature reading.
4. **Results vary between runs.** That same false alarm appeared in the evaluation but not in 3 later live runs of the folder. Rule-based checks are stable; AI judgments are not. Repeatability has not been measured formally. It is now a pilot gate.
5. **Retrieval adds little at this scale.** At 6 rules per query, retrieval missed SIG-001, the critical application signature rule, on 2 of 12 dev applications. Those documents were signed, so no result changed, but an unsigned one could have been missed. The fix raised retrieval to 10 and made signature rules always run. Retrieving 10 of 22 rules is close to reading the whole rulebook, so RAG's value must be proven on a much larger real rulebook.
6. **Some guardrails have not fired live.** No citation was rejected and no LOW band appeared. These paths are covered only by automated tests with a scripted model.
7. **Fail-closed has real evidence.** When the daily quota ran out during an aborted early run, 11 of 12 folders were marked Not Scored, and none received a false pass. That run's accuracy numbers are invalid and are not reported here.

Automated tests: 18 pass, using a scripted model. They verify logic, not AI accuracy.

9. Launch thresholds (go / no-go)

Gate to enter the pilot (offline, on the real golden set). All of these must hold: critical-defect recall ≥ 95% · overall recall ≥ 85% · precision ≥ 70% · better than the manual baseline on defects found · **0 fabricated evidence** in 100 audited findings · repeatability ≥ 95% · rule retrieval recall ≥ 98% · security and privacy sign-off.

Gate to scale. The pilot must show: analyst time −40% · usefulness ≥ 70% · rejection rate 10–40% · no audit submission containing a fabricated finding · a funded rulebook owner.

If AI judgments are unreliable but extraction works: keep AI extraction and rule-based checks, and leave judgment-heavy checks to people. That would be an "iterate" result, not a failure.

10. Lifecycle and drift

- **Rule drift:** rulebook changes are versioned, and each version gets a separate index.
- **Document drift:** new forms and layouts are found through rejected findings and low-confidence reads.
- **Model drift:** model versions are pinned, and the full evaluation runs before any change.

- **Regulatory change:** the compliance owner updates the rulebook; the model never infers new rules.
- **Quarterly** golden-set rerun, reported to the product owner.

06 · Business, Economics and Scaling

> **Important:** all figures here are **planning assumptions** for a hypothetical mid-size US property and casualty insurer. Folder reviews are done by onsite business analysts billed at **US$150 an hour**, and the same rate is used for every people role. No figure comes from a real company, and Phase 0 measurements should replace them. The working is shown so that any input can be changed. The main point: **at onsite rates, time savings alone pay back the investment in about 8–10 months. Reducing audit and regulatory risk is upside on top of that, not the core of the case.**

1. Monetization model

Internal value case. My decision is not to sell the product externally. The underwriting operations and compliance budgets would pay for it. Value is counted as onsite analyst hours no longer billed, audit rework avoided, and major remediation exercises avoided. Because onsite consultants are billed by the hour, hours saved can become cash savings directly, unlike salaried staff time.

Possible later option: if the Phase 4 platform covers several lines, run costs can be shared through an internal chargeback per business line, not per folder, so that usage is not discouraged.

2. Value logic

```mermaid flowchart LR A[Gaps found early
with evidence] --> BLess analyst time<b## 3. Assumptions

| # | Assumption | Value | Basis | Verify in |
|---|---|---|---|---|
| E1 | Commercial property folders reviewed per year | 12,000 | Assumption proposed with AI; not validated | Operations data |
| E2 | Analyst time per folder today | 45 minutes | Assumption proposed with AI; not validated | Phase 0 time study (30 folders) |
| E3 | Time reduction with the tool | 40% (conservative, = pilot target) / 50% (base) | Pilot target | Pilot |
| E4 | Loaded onsite rate, all people roles | US$150 / hour | My planning rate for onsite business analysts; applied to compliance experts, engineers and reviewers too | Vendor rate cards, finance |
| E5 | Audit findings needing rework per year | 200 | Assumption proposed with AI; not validated | Past audit reports |
| E6 | Rework per finding | 8 hours | Assumption proposed with AI; not validated | Pilot |
| E7 | Audit rework avoided | 30% | Assumption proposed with AI; not validated | Pilot |
| E8 | Cost of one major audit remediation exercise | US$150,000 (1,000 hours × US$150); excludes fines | Assumption proposed with AI; not validated | Compliance, finance |
| E9 | AI calls per folder | 15 | Measured about 14 per folder, rounded up | Pilot logs |
| E10 | Tokens per call | About 3,000 input + 500 output | Estimate, not measured | Pilot logs |
| E11 | API price | US$0.30 / M input, US$1.50 / M output | Illustrative; **check the provider's current price list** | Phase 1 procurement |
```

#	Assumption	Value	Basis	Verify in
E12	Working hours per FTE per year	1,700	Assumption proposed with AI; not validated	HR / finance

4. Cost drivers

One-time (Phase 0–1): ≈ US$360,000

Item	Hours × US$150	US$
Build: engineers and integration	1,200	180,000
Rulebook encoding with compliance experts	500	75,000
Golden set: expert labeling (see [05)	450	67,500
Security, privacy review, access control	250	37,500

Recurring (per year, after launch): ≈ US$178,000

Item	US$ / year	Note
Model / API cost (12,000 folders × about US$0.025)	**~300**	About 0.2% of run cost. At 10× the assumed price, about US$3,000
Hosting, storage, logging	5,000	Company cloud
Rulebook owner (0.5 FTE = 850 hours: rules, rejected findings, drift)	127,500	Largest cost
Quality review: golden-set refresh and evidence audits (300 hours)	45,000	The expert review that remains
Support and training	Not costed separately	Assumed to fit within the rulebook owner's role during the pilot. This needs testing, and scale would need its own budget

Insight: consumer AI products face heavy inference bills (the *OpenAI Scaling / Pricing* cases). Here **inference is a tiny cost, and people are almost all of it**: they keep rules current and check quality. A cheaper model would not change the economics, while a slower, more careful model could be worth many times the price if it improves accuracy.

5. Unit economics

Per folder	Conservative	Base
Run cost (US$177,800 ÷ 12,000)	US$14.82	US$14.82
Value of time saved (18 / 22.5 minutes at US$2.50 a minute)	US$45.00	US$56.25
Value of audit rework avoided (US$72,000 spread across all folders)	US$6.00	US$6.00
Value per folder (efficiency only)	**US$51.00**	**US$62.25**

Break-even: run costs are covered, after rework savings, once the tool saves about **3.5 minutes per folder** (8% of the assumed 45 minutes).

6. Annual business case and payback

Current annual cost of manual review: 12,000 × 0.75 hours × US$150 = **US$1.35 million**.

Scenario	Annual benefit	Net / year	Payback on US$360,000
Downside: time savings only −20%	US$342,000	US$164,200	~26 months
Efficiency only, conservative (−40%)	US$612,000	US$434,200	~10 months

Scenario	Annual benefit	Net / year	Payback on US$360,000
Efficiency only, base (−50%)	US$747,000	US$569,200	~8 months
Conservative + **1 avoided major remediation / year**	US$762,000	US$584,200	~7 months
Base + **1 avoided major remediation / year**	US$897,000	US$719,200	~6 months

Annual benefit = time saved + US$72,000 audit rework avoided (+ US$150,000 per avoided remediation). Net = benefit − US$177,800 run cost.

Decision implications

- On **efficiency alone, the investment is good**, paying back in about 8–10 months if the pilot reaches its 40% time target.
- The case is **sensitive to the time saving**: at only 20% it takes about 26 months. That is why the Phase 0 time study and the pilot's time measurement matter most.
- **Avoided remediation is upside.** The prototype suggests how it could happen, but **there is no evidence yet of how often it would**, so the case does not rely on it.
- Savings become cash only if billed onsite hours are actually reduced, or the onsite team does not grow with volume. The pilot should agree this with the vendor and finance up front.
- Regulatory penalties and renewal retention are not included. Both may matter, but cannot be estimated responsibly before the pilot.

s well as time. It should count serious gaps found by the tool that manual review missed. - Time released only creates cash value if capacity is actually redeployed or hiring is avoided. - Regulatory penalties and renewal retention are not included. Both may matter, but cannot be estimated responsibly before the pilot.

7. Scaling risks

Risk	Why it matters	Mitigation
Latency	33–42 seconds per folder suits one reviewer, not thousands of folders at once	Batch overnight in Phase 3; parallel calls; enterprise quotas
Rate limits and cost	The free tier stopped the first run at 20 requests a day	Paid enterprise capacity; budget alerts; never use a free tier for real data
Cost growth	More lines add folders and rules	API cost stays small; each line needs a rulebook owner, so each needs its own business case
Workflow complexity	Business lines use different document sets and naming	Taxonomy per line; begin standalone, integrate after the pilot
Governance burden	Sensitive data, audit expectations, model change control	Access controls, logs from day one, regression tests before changes
Document quality	Poor scans reduce accuracy	Route poor scans to people; improve scan standards at intake
Adoption	Analysts may fear replacement or distrust findings	Position as an assistant; involve analysts in shadow mode; do not use logs for performance scoring
Error at scale	One wrong rule can produce thousands of wrong flags	Rule change testing; staged rollout; no bulk scan until accuracy is proven

8. Operations implications

- **New role:** a rulebook owner in compliance at 0.5 FTE. Without one, rules go stale and the product should not launch.
- **Analyst work changes:** time moves from reading to deciding findings, so analysts need training on when to reject or escalate.

- **Support:** first-line help comes from the rulebook owner during the pilot. Scale needs a named support route, which is not yet costed.
- **Escalation capacity:** compliance and underwriting experts must have time for escalated findings, or the queue simply moves to them.
- **Monthly and quarterly work:** rejected findings are reviewed monthly, and the golden set is rerun quarterly.
- **Change control:** model, prompt and rule changes all need an evaluation run before release.

07 · Ethics, Governance and Risk Note

PolicyGuard AI handles sensitive policyholder data, and its output can influence audit submissions and renewals. It does not price risk or decide eligibility, but a wrong finding can still affect a customer's renewal. The main risks are **missed defects, false evidence presented with confidence, data exposure and over-reliance**. The full register with owners is in Appendix/Risk_Note.

This note gives product-level context. It is not legal advice. The setting is a hypothetical US insurer. Any reference to state insurance regulation, NAIC guidance on insurers' use of AI, or privacy law such as the Gramm-Leach-Bliley Act must be confirmed by the insurer's legal and compliance teams.

1. Main risks

#	Risk	Example	Likelihood	Impact
R1	**Missed defect / false pass**	An unsigned application passes, and the auditor finds it. *Found in testing:* retrieval skipped the signature rule on 2 applications	Medium	**Very high**
R2	**Hallucination: fabricated or wrong evidence**	A finding quotes text that the document does not contain	Low–Medium	**High**
R3	**False alarm on a real signature**	*Found in testing:* a valid scanned signature was described as "a squiggle"	Medium	Medium
R4	**Inconsistent results between runs**	*Found in testing:* one folder's findings changed between runs	**High**	Medium
R5	**Over-reliance / automation bias**	A reviewer accepts every finding, or treats a Pass as a sign-off. *Found in testing:* the first UI pre-selected "Approve renewal"	Medium	**Very high**
R6	**Personal and financial data exposure**	Policyholder documents sent to a provider that retains them, or seen by the wrong staff	Low–Medium	**Very high**
R7	**Bias: unequal customer impact**	Older or scanned folders from certain regions get more flags, delaying those customers	Medium	Medium–High
R8	**Silent failure**	*Found in testing:* the first prototype gave 100/100 and "Approved for Automated Renewal" when AI analysis failed	Fixed; Low	**Very high**
R9	**Outdated rules**	A regulation changes, but the rulebook does not	Medium	High
R10	**Misuse**	Audit logs used to rate individual analysts; the score used as an official compliance certificate	Low–Medium	Medium

2. Harm scenarios (who could be harmed, and how)

- **Policyholders:** a renewal is delayed by a false flag, or personal data is exposed (R3, R6, R7).
- **The insurer:** an audit report contains a false or incomplete finding, leading to regulatory and reputational damage (R1, R2, R5).
- **Analysts and reviewers:** blamed for a decision influenced by AI, or judged using logs they did not expect to be used that way (R5, R10).
- **Compliance officers:** they sign reports that they cannot fully explain (R2, R4).

3. Controls

Guardrails (in the product)

Control	Addresses	Status
AI **never approves**; all renewal outputs are suggestions	R5	**In prototype**

Control	Addresses	Status
Fail closed: any failed analysis → Not Scored, no recommendation, and a warning	R1, R8	**In prototype; confirmed in a real quota outage**
Any open finding blocks a Pass	R1	**In prototype**
Signature rules always run, regardless of retrieval	R1	**In prototype**
Quote check: quotes are compared with document text; a quote not found gets LOW confidence	R2	**In prototype**
Scans are marked "no text layer" and cannot reach HIGH confidence	R2, R3	**In prototype**
Retrieved-rule check: AI cannot cite a rule it was not given	R2	**In prototype**
AI rationale checked for valid rule IDs and rejected if it says "approved"	R2, R5	**In prototype**
Required reason per finding; renewal decision has no default; sign-off blocked until complete	R5	**In prototype**
Synthetic data only, with a notice in the UI	R6	**In prototype**
Repeatability measured and thresholded	R4	Pilot
Access control and masking of personal data	R6	Pilot
Contract terms: **the provider may not train on or retain the data, and it stays in an approved region**; in-house model as fallback	R6	Pilot prerequisite
Tamper-evident audit log (today's is a plain file)	R2, R5	Pilot
Rule versions with an accountable owner	R9	Pilot
Flag-rate check by branch, era and customer segment	R7	Pilot

Review process

- **Each finding** is decided by a person, with a reason (in prototype).
- **Monthly** review of rejected findings and escalations by the rulebook owner.
- **Quarterly** golden-set rerun, plus a 100-finding evidence audit.
- **Before any model, prompt or rule change:** the full evaluation must pass.

Escalation path

Ambiguous or high-risk finding → compliance or underwriting SME. Suspected wrong finding → rulebook owner within 2 working days. Fabricated evidence or a tool-caused critical miss in an audit submission → **product owner returns the product to shadow mode** until the root cause is fixed. Suspected data incident → the insurer's security and privacy process.

Logging / auditability

The prototype logs the reviewer, time, folder, AI score, AI recommendation, each human decision and reason, and the renewal decision. The audit report includes the findings and evidence. The pilot adds model and rulebook versions, and tamper protection. **Logs improve the product and support audits. They are not for rating individuals.** This goes into the pilot charter before launch (R10).

4. Launch blockers

Any one of these stops the launch, or pauses a running pilot:

1. **Any fabricated evidence** in the 100-finding audit.
2. Recall on critical defects below 95% on the real golden set.
3. Any case of a failed analysis giving a score, Pass or recommendation.
4. Repeatability below 95%.

5. **No security and privacy approval** for sending documents to the chosen provider. Fallback: an in-house model, which means re-planning.
6. Access control cannot limit who sees policyholder data.
7. During the pilot: a rejection rate above 40% (not useful), or below 5% without outcomes to support it (findings possibly accepted unread). A rate of 5–10% is investigated.
8. Any audit submission or renewal decision relying on AI output without recorded human review.

5. Launch checklist

Each item needs a named owner and a recorded yes before the pilot goes live.

Area	Check
Evaluation	All offline gates in Part 05 are passed on the real golden set
Review	Analysts are trained; every finding needs a recorded human decision and reason; 100-finding evidence audit is complete
Privacy and security	Privacy approval given; enterprise terms signed; access control and masking tested
Rollback	Return to shadow mode has been tested; the product owner can trigger it; analysts can revert to manual review the same day
Escalation	SME and rulebook owner are named; the 2-working-day route for wrong findings is live; the data-incident route is known
Monitoring	Rejection rate, usefulness and critical misses are reported weekly during the pilot
Governance	Model and rulebook versions are pinned; the pilot charter says logs are not used to rate individuals

6. Claims policy

- **Allowed:** *"PolicyGuard AI helps reviewers find folder gaps earlier, shows its evidence, and says when it could not complete an analysis."*
- **Not allowed:** "AI certifies compliance", "automated renewal", "X% automation", or any synthetic-data accuracy figure presented as real-world performance.